Lec 32 Minimal Coupling

Minimal number of interactions between charged particle and an EM field.

Contrast w/ Pauli coupling which includes magnetic moment explicitly ($\hat{H} = -\vec{\mu}\cdot\vec{B}$)

Recall from EM: $= 0$ → we will assume time ind.

$$\vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t} \; ; \quad \vec{B} = \vec{\nabla}\times\vec{A}$$

Multiple $\vec{A}$ and ϕ generate the same $\vec{E}$ and $\vec{B}$.

$$\hat{H} = \frac{(\hat{p} - e\vec{A}(\hat{x}))}{2m} + e\phi(\hat{x})$$

$$\hat{P} \longrightarrow \hat{\pi} = \hat{p} - e\hat{A}$$

Classically $L = \frac{m\dot{x}^2}{2} - e\phi + e\dot{x}\cdot\vec{A}$

Euler-Lagrange $m\ddot{x} = e(\vec{E} + \dot{x}\cdot\vec{B})$

canonical momentum $p = \frac{\partial L}{\partial \dot{x}} = m\dot{x} + e\vec{A}$

So $H = p\cdot\dot{x} - L = \frac{m\dot{x}^2}{2} + e\phi$

BTW $\exp(iS) \supset \exp\left(ie\int \vec{A}\cdot d\vec{x}\right)$

commutation: $(\hat{p} - e\hat{A})^2 = \hat{p}^2 - e(\hat{p}\cdot\vec{A} + \vec{A}\cdot\hat{p}) + e^2\vec{A}^2$

(Is it Hermitian?)

Recall the identity $[\hat{P}_i, F(\hat{x})] = -i\partial_i F(\hat{x})$

Therefore: $p\cdot A - A\cdot p = -i\nabla\cdot A \longrightarrow 0$ in Coulomb gauge.

Example $\vec{A} = \frac{\vec{B}\times\hat{x}}{2}$; ($B = \nabla\times A$ and $\nabla\cdot A = 0$)

$$\vec{A}\cdot\hat{p} = \frac{(B\times\hat{x})}{2}\cdot\hat{p} = \frac{B\cdot(\hat{x}\times\hat{p})}{2} = \frac{\vec{B}\cdot\vec{L}}{2}$$

$$\Rightarrow \quad \hat{H} = \frac{\hat{p}^2}{2m} - e\frac{\vec{B}\cdot\vec{L}}{2m} + \frac{e^2}{2m}(B\times\hat{x})^2 + e\phi$$

$-\hat{\mu}\cdot\vec{B}$ w/ $\hat{\mu} = \frac{e}{2m}\hat{L}$ → Diamagnetic small for small B.

Time evolution:

$$\frac{d\hat{x}_i}{dt} = \frac{[\hat{x}_i, \hat{H}]}{i} = \frac{\hat{p}_i - eA_i}{m} \Rightarrow x(t) = \frac{\hat{\pi}}{m}t + \hat{x}(0)$$

(note $[\hat{x}_i, \hat{\pi}_j] = i\delta_{ij}\hat{\mathbb{1}}$)

$\hat{p}$ → canonical momentum

$\hat{\pi}$ → kinetical momentum

$$\hat{H} = \frac{\hat{\pi}^2}{2m} + e\phi(x)$$

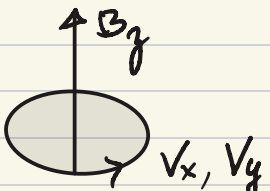
Note $[\hat{p}_i, \hat{p}_j] = 0$, but

$$[\hat{\pi}_i, \hat{\pi}_j] = i\,e(\partial_i A_j - \partial_j A_i) = i\,\varepsilon_{ijk}\,e\,B_k$$

bc/ $B_k = \varepsilon_{klm}\partial_l A_m$

or $\vec{B} = \vec{\nabla}\times\vec{A}$

$\hat{V}_i = \frac{\hat{\pi}_i}{m}$ incompatible with $\hat{V}_j$.



BTW: Lorentz Force comes out of our formalism.

$$\frac{d\hat{\pi}_i}{dt} = \frac{[\hat{\pi}_i, \hat{H}]}{i} + \frac{\partial\hat{\pi}_i}{\partial t}$$

$$= \frac{e}{2m}\varepsilon_{ijk}\left(\hat{\pi}_j B_k + \hat{B}_k\hat{\pi}_j\right) - e\partial_i\phi - e\frac{\partial A_i}{\partial t}$$